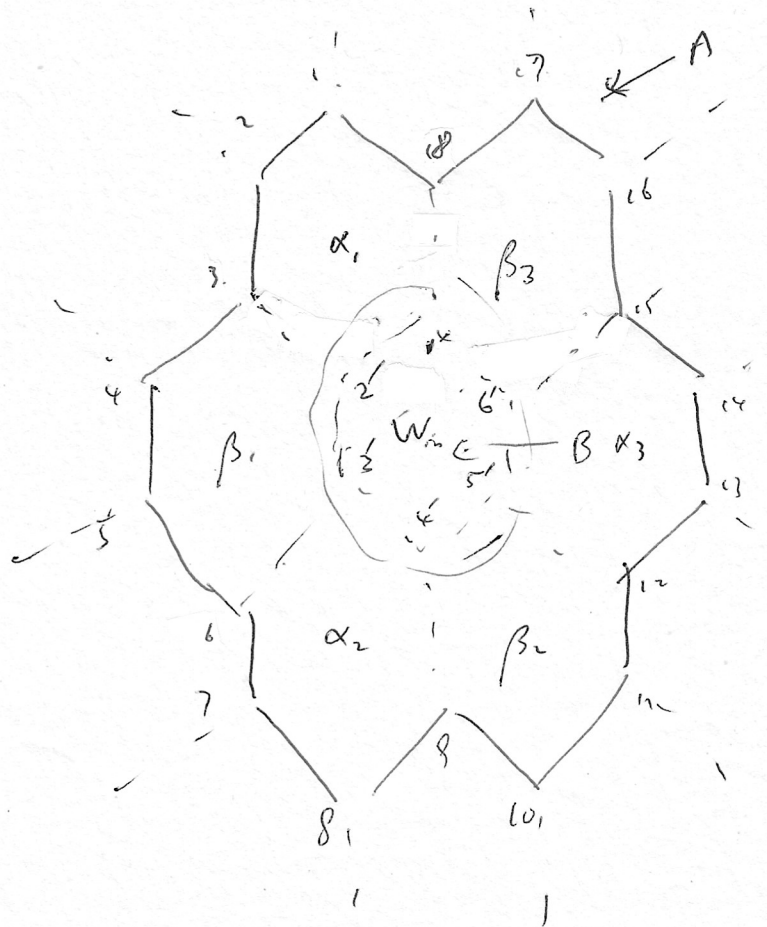


Now we consider other examples. (cf. CC's)

A has 18 sites
 B has 6 sites
 AB has 24 sites
 $L_{out} = 6$
 $L_{in} = 3$



$$P_{AB} = \frac{1}{2^5} \underbrace{\left(\frac{1+W_{\alpha_1}}{2} \right) \dots \left(\frac{1+W_{\beta_3}}{2} \right) \left(\frac{1+W_{in}}{2} \right)}_{6 \text{ factors}} \left\{ \frac{1}{2^{12}} [1 + \dots] \right\}$$

$\uparrow$
 $P_{AB,F}$
 $\dots (8.1)$

$W_{\alpha_1}, W_{\alpha_2}, W_{\alpha_3}, W_{\beta_1}, W_{\beta_2}, W_{\beta_3}, W_{in} = W$ entirely in A.
 and is the only closed loop in A.

$$\rho_A = \text{Tr}_B \hat{\rho}_{AB}$$

Following strategy in § 8, we write

half of all w_α and their products

$$\hat{\rho}_A = \frac{1}{2^7} \left(\frac{1+W}{2} \right) \frac{1}{2^{10}} \frac{1}{2^6} \text{Tr}_B \left(1 + \overbrace{w_{\alpha_1} \cdots w_{\alpha_n} + \text{products}}^{\text{half of all } w_\alpha \text{ and their products}} \right) 1$$

$$\left(1 + (\text{dimers of } (A \cup B)) \right)_2 \}$$

... (8.2)

and seek finite trace.

First, concentrate on 1 and operators that will not involve dimers within A

$$1_1 = 1_2$$

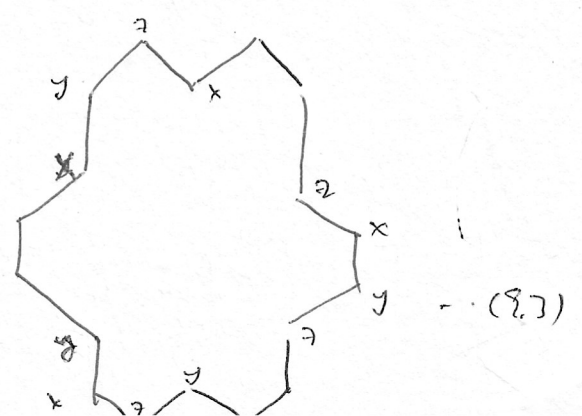
$$(w_{\alpha_1} w_{\alpha_2} w_{\alpha_3}) \text{ from } 1 \quad \left((\sigma_2^z \sigma_3^z) (\sigma_4^x \sigma_5^x) (\sigma_6^y \sigma_1^y) \right) \text{ from } 2$$

note $(w_{\alpha_1} (\sigma_6^y \sigma_1^y)) \equiv w'_{\alpha_1}$ entirely in A etc

get

$$\langle (\sigma_2^z \sigma_3^z) (\sigma_4^x \sigma_5^x) (\sigma_6^y \sigma_1^y) \rangle w'_{\alpha_1} w'_{\alpha_2} w'_{\alpha_3}$$

(not dimer in A)



$(W_{\beta_1} W_{\beta_2} W_{\beta_3})$ from 1 $\times$ (same term in 2)

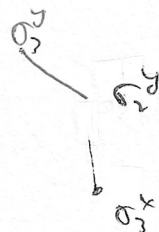
actually just double counts

2! $W(W_{\alpha_1} W_{\alpha_2} W_{\alpha_3})$ already included in (9.2).

• These are the only terms.

eg. If we include both $W_{\alpha_1} W_{\beta_1}$ then we must include σ_2^y , which forces us to include it as part of a string.

but then there is a leftover σ_3^y



• including a full string which ends back in A

must be equivalent to a direct string entirely in A

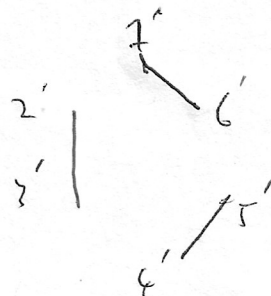
after multiplying by the W 's

(amounts to moving strings on AB to A)

The set $\sigma_2^z \sigma_3^z \sigma_4^x \sigma_5^x \sigma_6^y \sigma_7^y = \bar{P}$... (9.4)

can actually be viewed as three disconnected dimer operators

(and can be understood as a special dimerized limit of $\mathbb{Z}_2$ with a Kekulé pattern rather than $\pi\pi$)



Expected structure of $\hat{\rho}_A$:

$$\hat{\rho}_A = \frac{1}{2^7} \left(\frac{1+W}{2} \right) \frac{1}{2^{10}} \left[1 + (W'_1 W'_2 W'_3) \langle \bar{P} \rangle \right.$$

+ (dimers in A) +

$$\left. + \left[(W'_1 W'_2 W'_3) (\text{dimers in A}) \right] \right]$$

$\langle \bar{P} + \text{dimers in A} \rangle$

... (9.5)

eg $\langle \sigma_2^z \sigma_3^z \rangle \sigma_2^z \sigma_3^z$

eg $(W'_1 W'_2 W'_3 \sigma_2^z \sigma_3^z) \langle \sigma_2^z \sigma_3^z \bar{P} \rangle$

This last set of operators can look unfamiliar.

and contain the term generated by Ω of C .

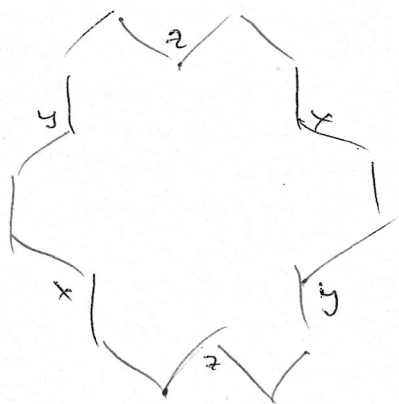
$$\Omega \text{ of } C \equiv \underbrace{(\sigma_{18}^z \sigma_3^y \sigma_6^x \sigma_9^z \sigma_{12}^y \sigma_{15}^x)}_{A's} \underbrace{(\sigma_{1'}^z \sigma_{2'}^y \sigma_{3'}^x \sigma_{4'}^z \sigma_{5'}^y \sigma_{6'}^x)}_{W_{in}} \quad \leftarrow B \quad \dots (9.6)$$

$$\text{coef. } \langle \Omega \rangle = \langle \sigma_{18}^z \dots \sigma_{15}^x \sigma_{1'}^z \dots \sigma_{6'}^x \rangle = \langle (\sigma_{18}^z \dots \sigma_{15}^x) \rangle$$

so finite trace with W_{in} factor 1 in (9.2)

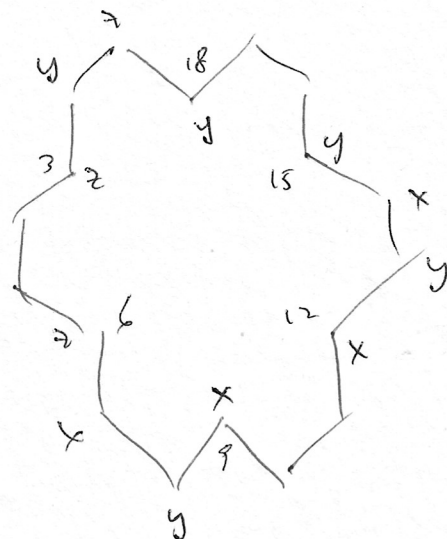
$$\langle (\sigma_{18}^z \sigma_{1'}^z) \dots \rangle \quad \pi \quad 0$$

this results in



$$= (W'_{\alpha_1} W'_{\alpha_2} W'_{\alpha_3})$$

↑
(see (9.3))



2nd factor is product of three linear operators in A .

$$\begin{aligned} & \sigma_{18}^y \sigma_{1'}^z \sigma_{2'}^y \sigma_{3'}^z \quad \leftarrow A \text{ handle} \\ & \sigma_6^z \sigma_7^x \sigma_8^y \sigma_9^x \quad \leftarrow \\ & \sigma_{12}^x \sigma_{13}^y \sigma_{14}^x \sigma_{15}^y \quad \checkmark \end{aligned}$$

∴ This operator is thus like

$$(w_{\alpha_1} w_{\alpha_2} w_{\alpha_3}) \underbrace{(ic_{18} c_5)(ic_6 c_9)(ic_{10} c_{13})}_{\text{three strings in A}}$$

can be expressed in terms of Majorana in AUB:
coefficient of this operator from (8.6)

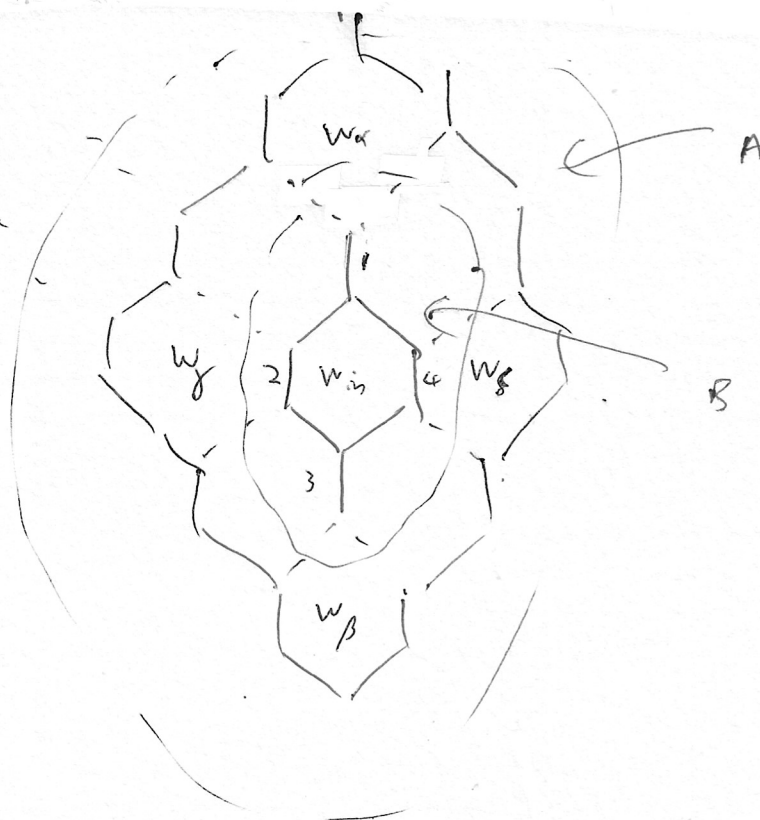
$$\langle (ic_{18}^2 c_{11}^2) \dots (ic_{15} c_{61}) \rangle$$

(pair of Majorana's)
product of 6 dimers involving both A & B

$$\sim \langle \sigma_{18}^z \dots \sigma_{15}^x \rangle \quad \text{just sites on A.}$$

Other examples:

(MS9.7)

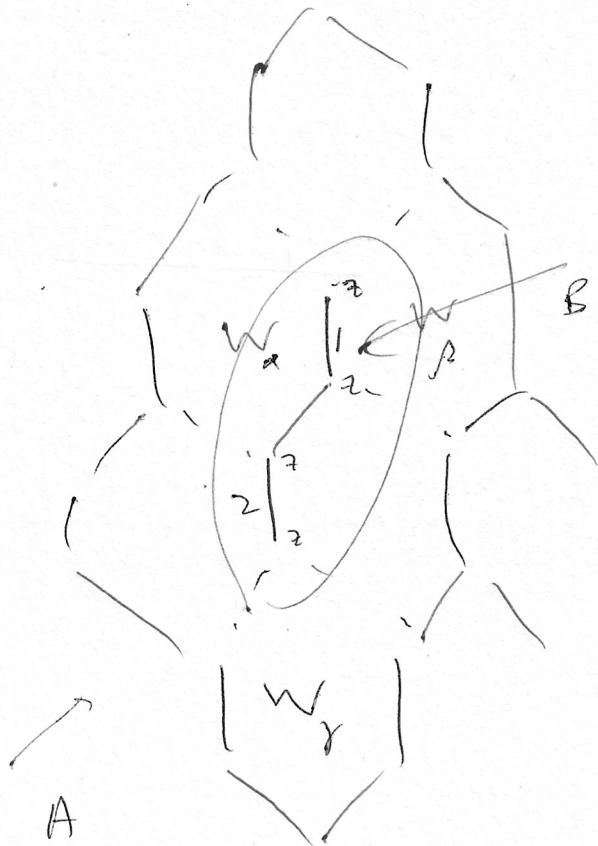


$$\hat{\rho} = \left[1 + \frac{(w_a w_b w_y w_s w_s w_{in}) (\sigma_{1a}^z \sigma_{1b}^z) (\sigma_{2a}^z \sigma_{2b}^z) (\sigma_{3a}^z \sigma_{3b}^z) (\sigma_{4a}^z \sigma_{4b}^z)}{\langle (\sigma_{1a}^z \sigma_{1b}^z) (\sigma_{2a}^z \sigma_{2b}^z) (\sigma_{3a}^z \sigma_{3b}^z) (\sigma_{4a}^z \sigma_{4b}^z) \rangle} + \dots \right]$$

= $w_a' w_b' w_y' w_s'$ entirely in A

$$\hat{\rho}_A = \frac{(1 + w_{in})}{2} \left[1 + (w_a' w_b' w_y' w_s') \langle P \rangle + (\text{dim in A}) + (w_a' w_b' w_y' w_s') (\text{dim in A}) \times \langle (\text{dim in A}) \cdot P \rangle \right]$$

(9.7)

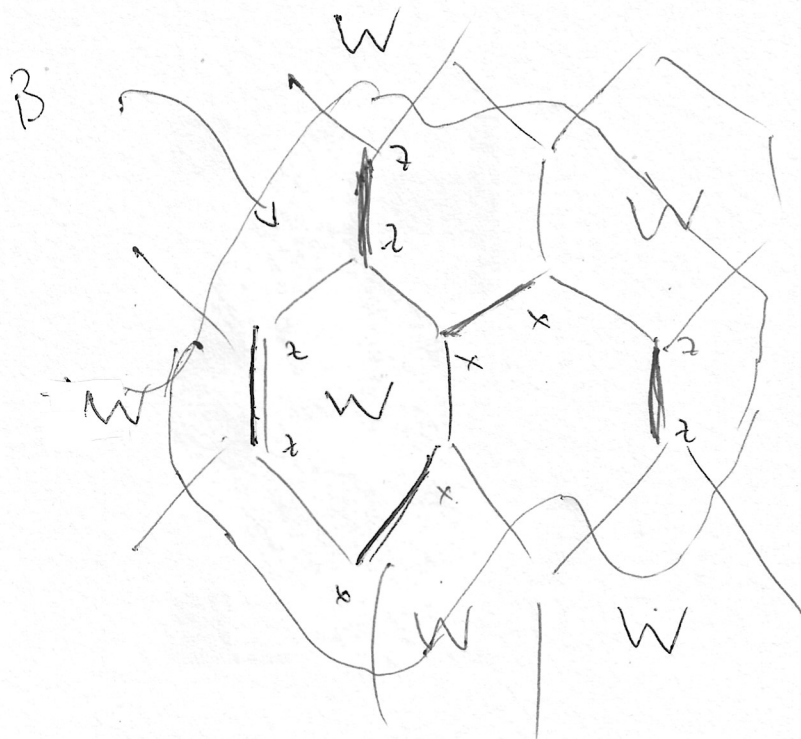


we note

$$\underbrace{(W_\alpha W_\beta W_\gamma) (\sigma_{\alpha\beta}^2 \sigma_{\beta\gamma}^2) (\sigma_{\gamma\alpha}^2 \sigma_{\alpha\beta}^2)}_{W'_\alpha W'_\beta W'_\gamma} \text{ is entirely in } A.$$

so

$$\hat{\rho} = \left[1 + \underbrace{W'_\alpha W'_\beta W'_\gamma}_{\substack{\text{in} \\ P}} < \underbrace{(\sigma_{\alpha\beta}^2 \sigma_{\beta\gamma}^2) (\sigma_{\gamma\alpha}^2 \sigma_{\alpha\beta}^2)}_{\substack{\text{in} \\ P}} \right]$$



the extra operator = $\underbrace{(\text{explicit } W) (\text{explicit dimer on } B)}$

note all edges involving B cancel

coef of this operator is $\langle \text{explicit dimer on } B \rangle$

again like "fermi parity of region B"

At least, when B has even # of sites, the above seems to apply generally.